3AL Tutorial 2 2024

1. Let G be a group of order 105. List all possible orders of elements of G.

2. Let p and q be distinct primes and G a group with order pq. Prove that every proper subgroup of G is cyclic.

   NOTE: We call a subgroup "proper" if it does not equal the group itself.

3. Let G be a group with subgroups H and K. Prove the following statements:

   (a) $H \triangleleft G$ if and only if H is the kernel of a homomorphism.

   (b) If H and K are normal in G, then $H \cap K \triangleleft G$.

   (c) If $H \triangleleft G$, then $H \cap K \triangleleft K$.

   (d) If $H \triangleleft G$ and $H \subseteq K$, then $H \triangleleft K$.

4. Let G be a group with subgroup H. Prove that if $|G : H| = 2$, then $H \triangleleft G$.

5. Let G be a group. Prove the following:

   (a) If $G/Z(G)$ is cyclic, then G is abelian.

   HINT: Let the generator of $G/Z(G)$ be denoted by gH (where $H = Z(G)$). Show that every element of G can be written as $g^n z$ for $n \in \mathbb{Z}$ and $z \in Z(G)$.

   (b) If G has order pq (for primes p and q, not necessarily distinct), then either G is abelian or $Z(G) = \{1\}$.

6. Let $G = \mathbb{Z}_4 \times \mathbb{Z}_4$ and $H = \{(0,0), (2,0), (0,2), (2,2)\}$ (here we drop the equivalence class notation $[\,]$ for ease of notation). Is G/H isomorphic to $\mathbb{Z}_4$ or K_4?

7. Let G be a group.

   (a) Prove that $\mathrm{Aut}(G)$, the set of automorphisms $\varphi : G \to G$, is a group under composition.

   (b) Let $x \in G$. Prove that the map $\sigma_x : G \to G$ defined by $\sigma_x(g) = xgx^{-1}$ is an automorphism.

(c) Let $x \in G$. Prove that $(\sigma_x)^{-1} = \sigma_{x^{-1}}$.

(d) Prove that $\mathrm{Inn}(G) = \{\sigma_x : x \in G\}$ is a subgroup of $\mathrm{Aut}(G)$.

(e) Find a group G such that $\mathrm{Inn}(G) \cong \{e\}$ (the trivial group consisting only of an identity element).

Extra problems

(I) Let G be a group with $H \triangleleft G$. Prove that if $|G : H| = n$ (where $n \in \mathbb{Z}^+$), then $g^n \in H$ for all $g \in G$.

(II) Show, via counterexample, that the image of a normal subgroup is not necessarily normal. That is, find groups G and G' with homomorphism $\varphi : G \to G'$ with $H \triangleleft G$ such that $\varphi(H) \ntriangleleft G'$.